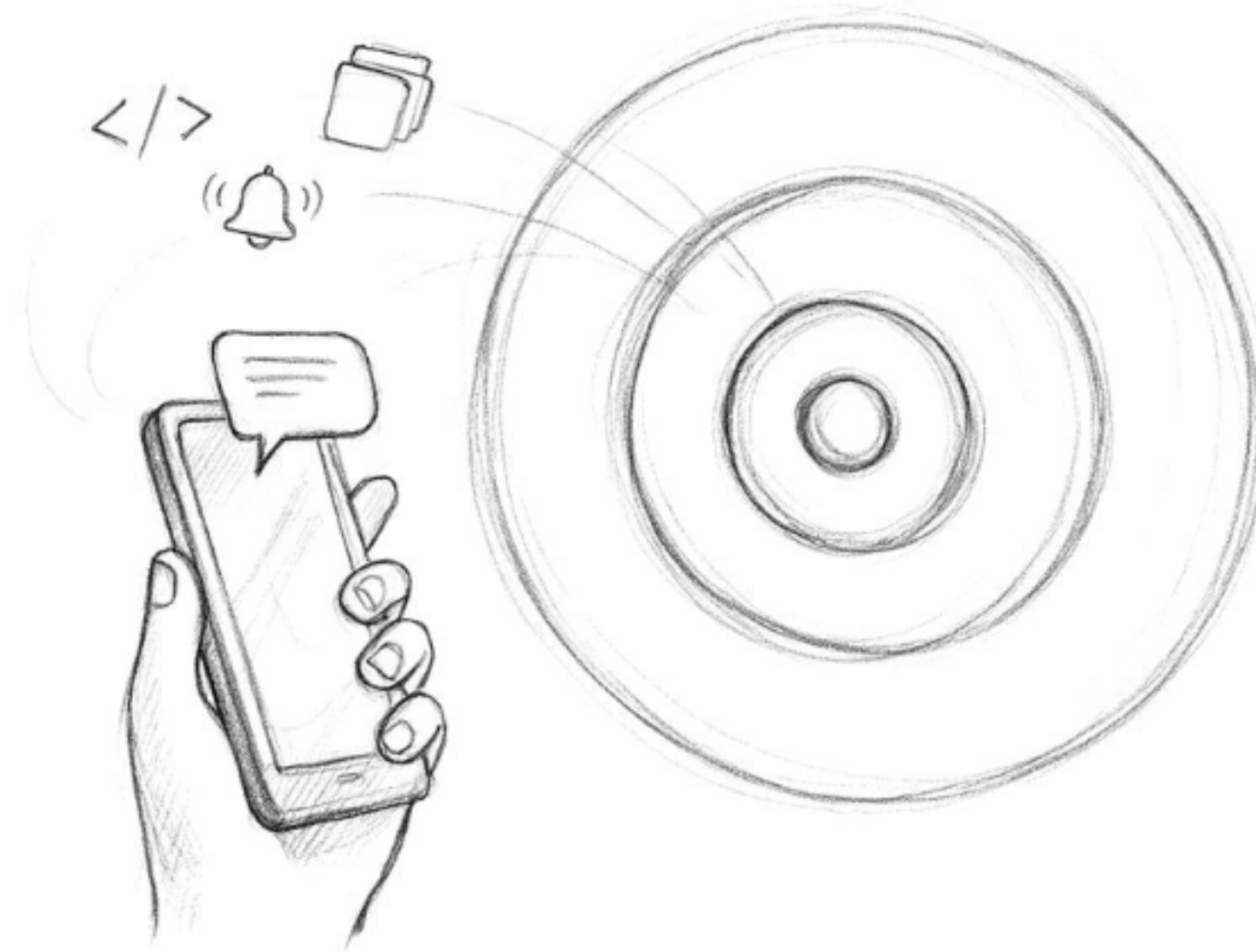


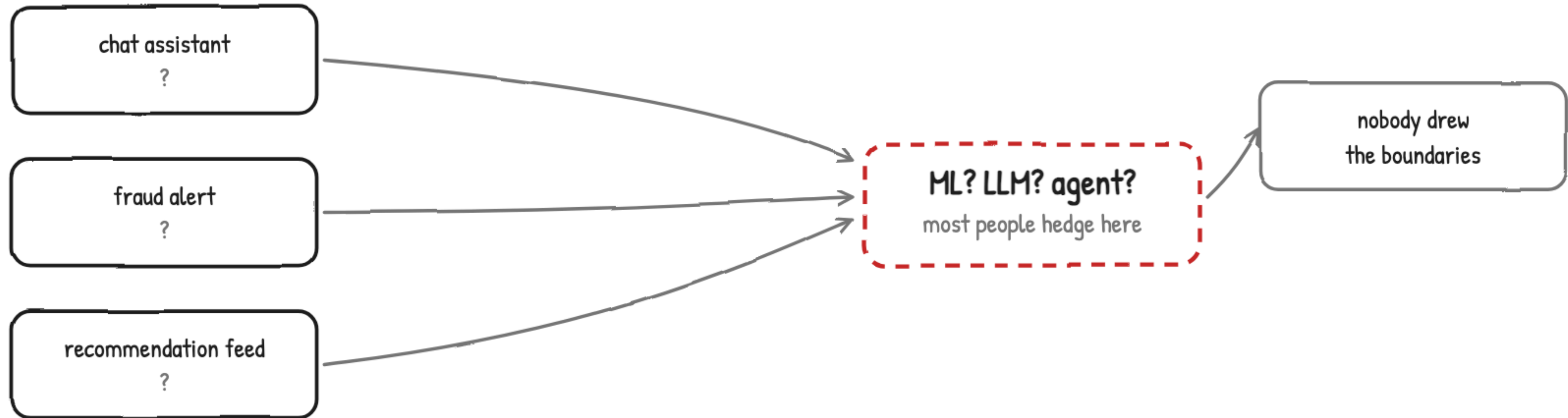
START HERE — The AI Map



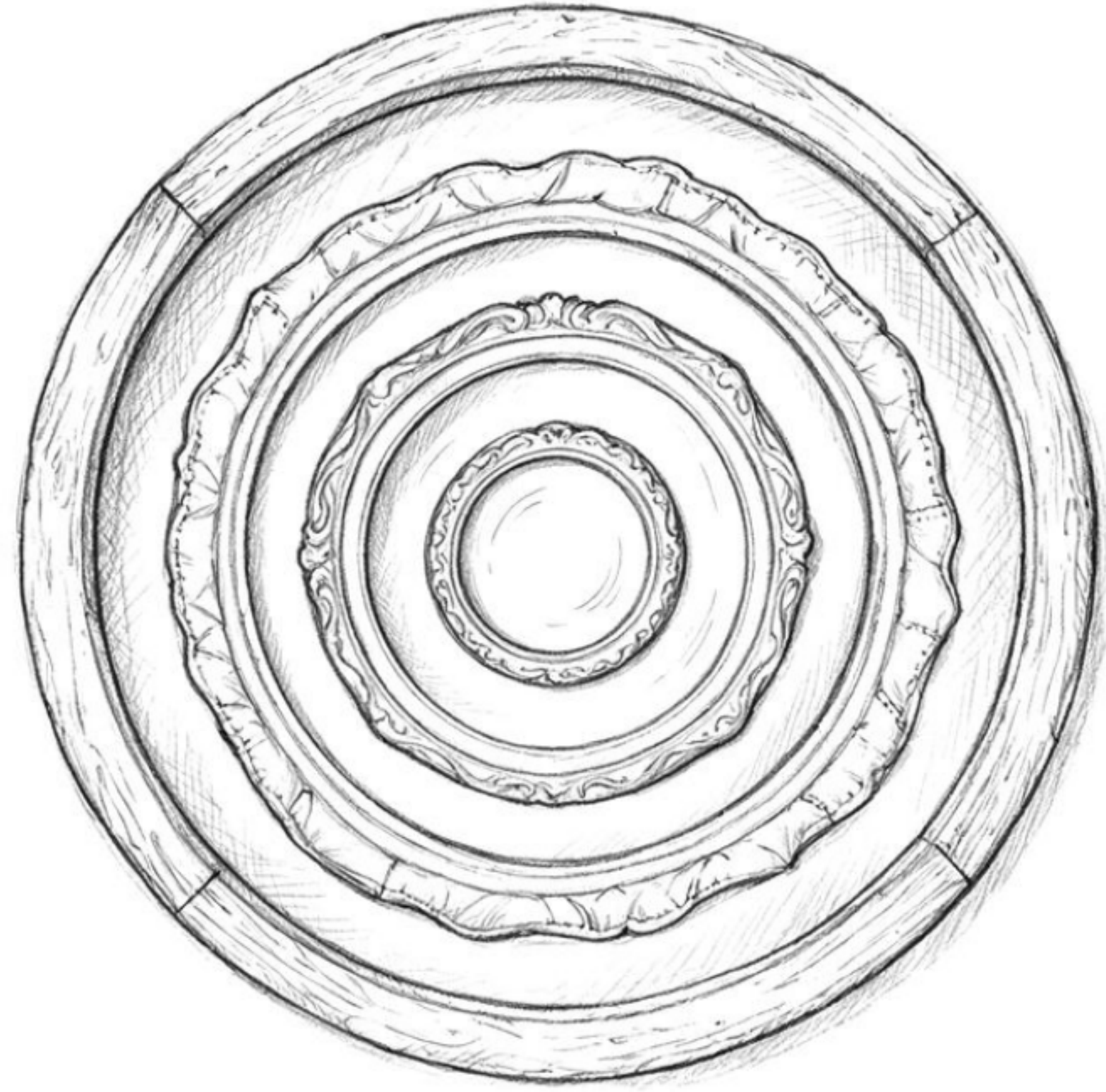
ML → Deep Learning → LLMs → Agents

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Every AI product hides one honest question

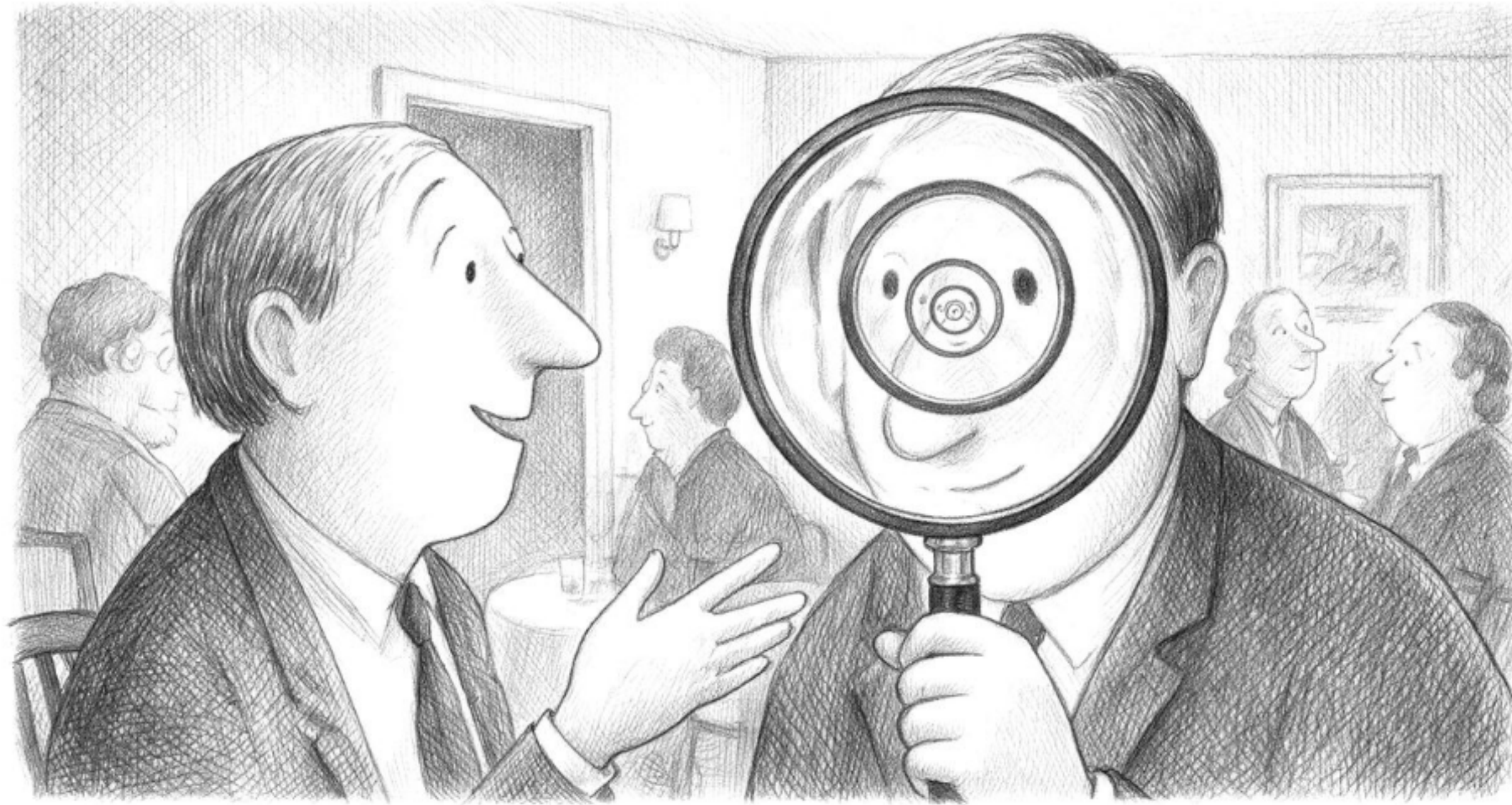


Is it ML, an LLM, or an agent?



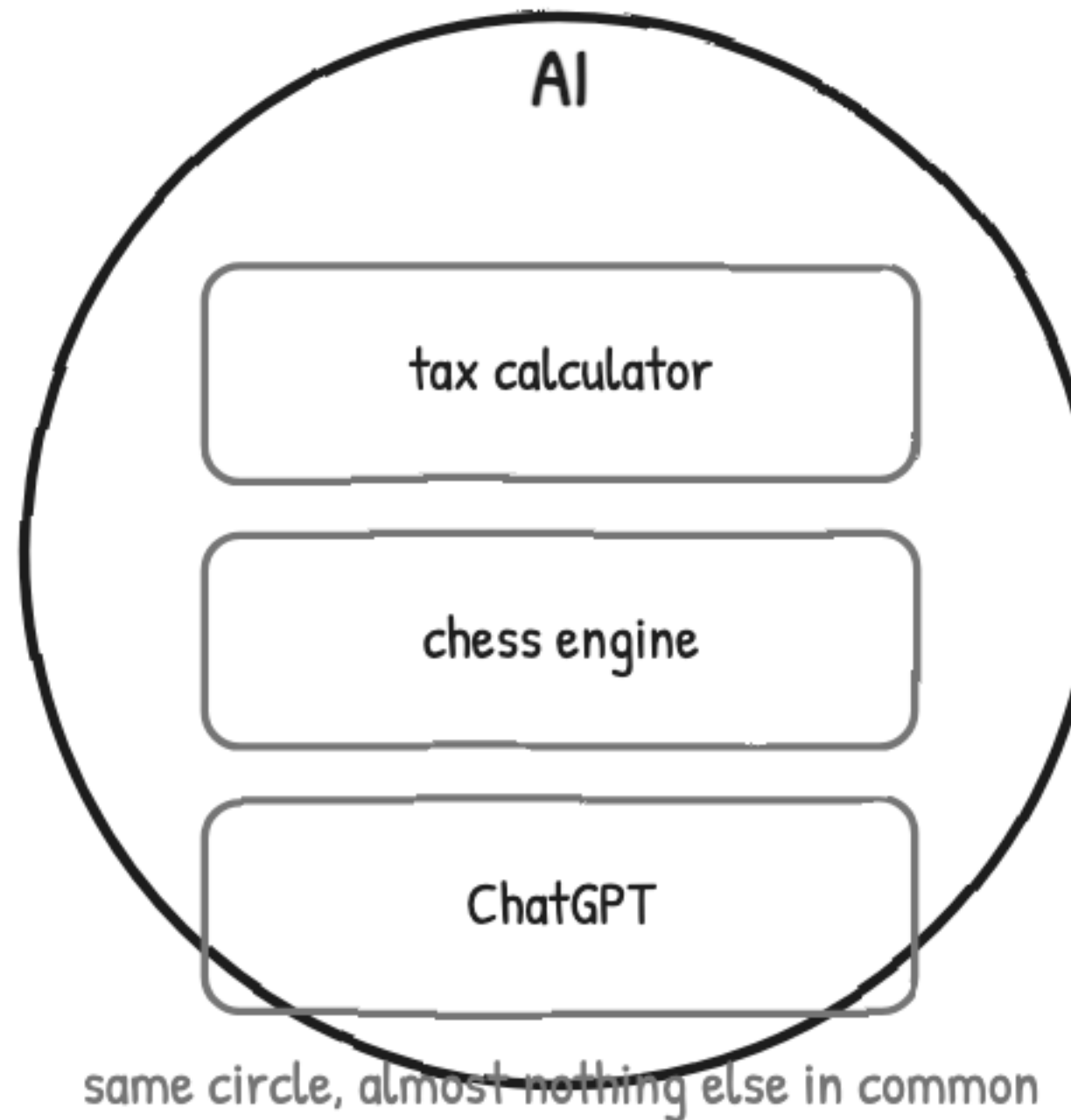
The map is nested, not a list

You already zoom levels when you describe your job



Nobody's wrong — just a different distance

AI is the widest circle — a goal, not a method

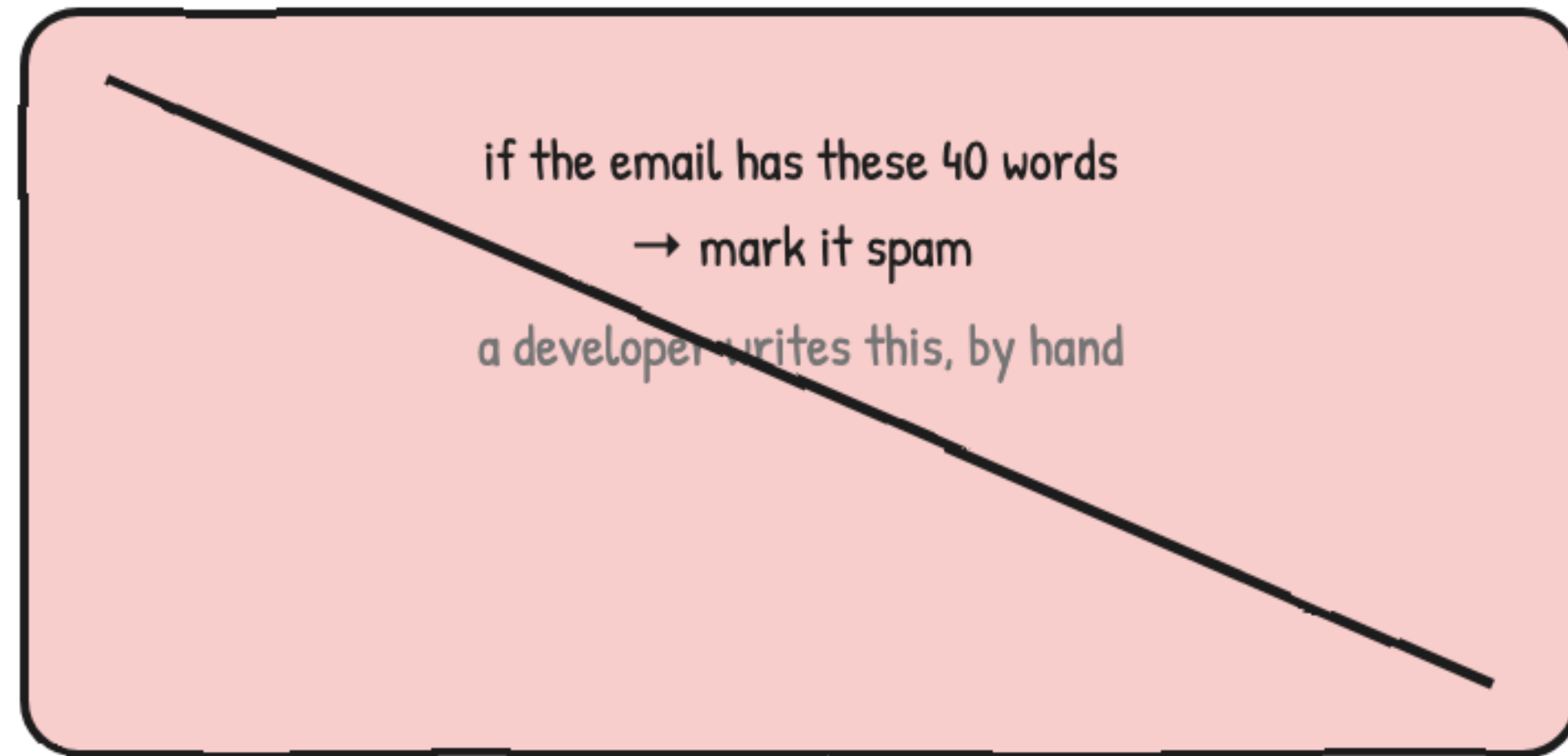


Anything that looks like it needs judgment

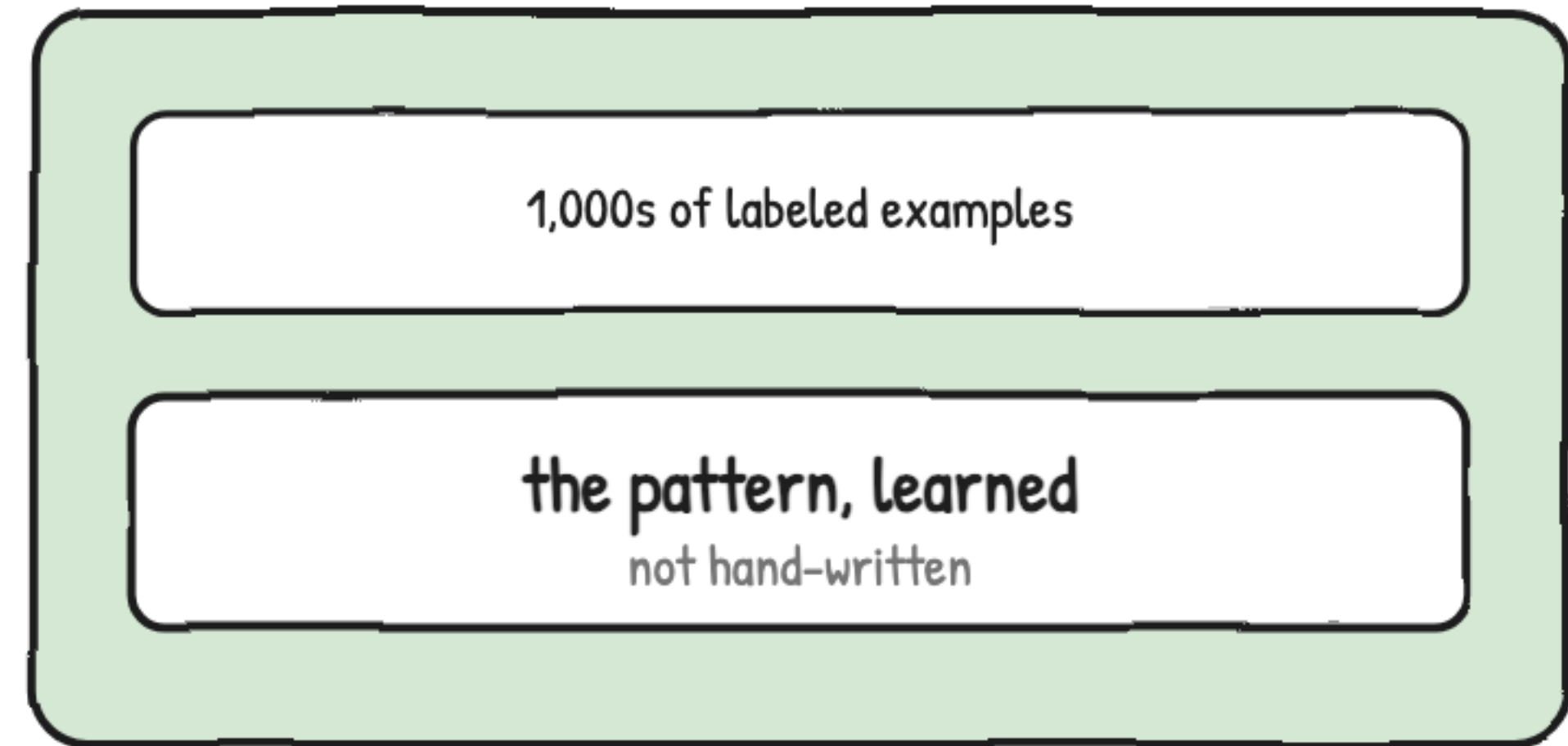
ML swaps hand-written rules for rules learned from data



the old way

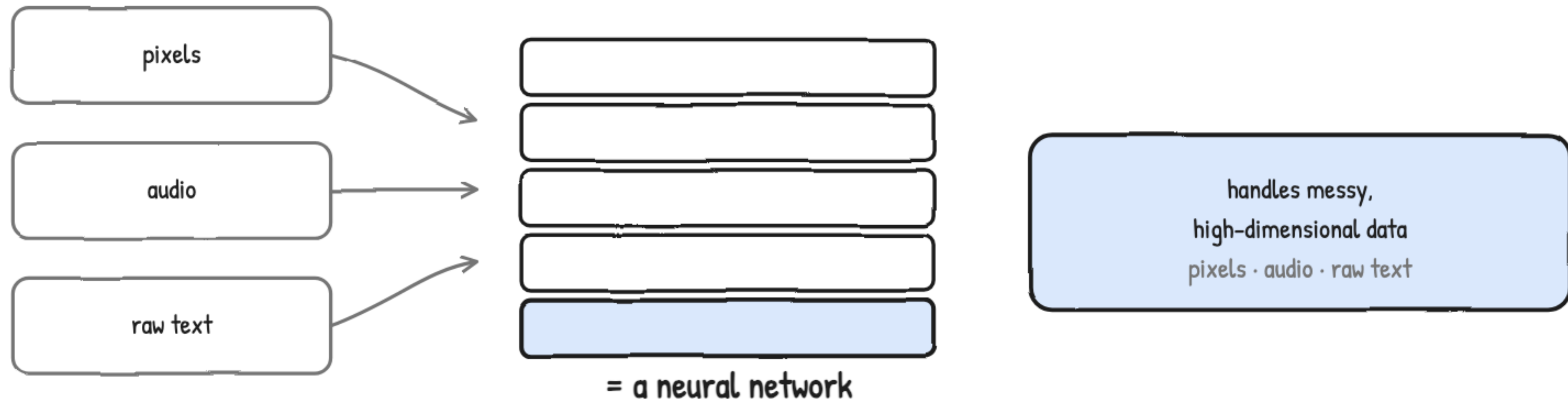


the ML way



The rules come from data, not an if/else tree

DL stacks layers to handle messy, high-dimensional data



Many small transformations, stacked

LLMs do one narrow thing extremely well, at huge scale



AI $\supset$ ML $\supset$ DL $\supset$ LLM

the whole internet
(all the text)

trained on

predict the next
chunk of text

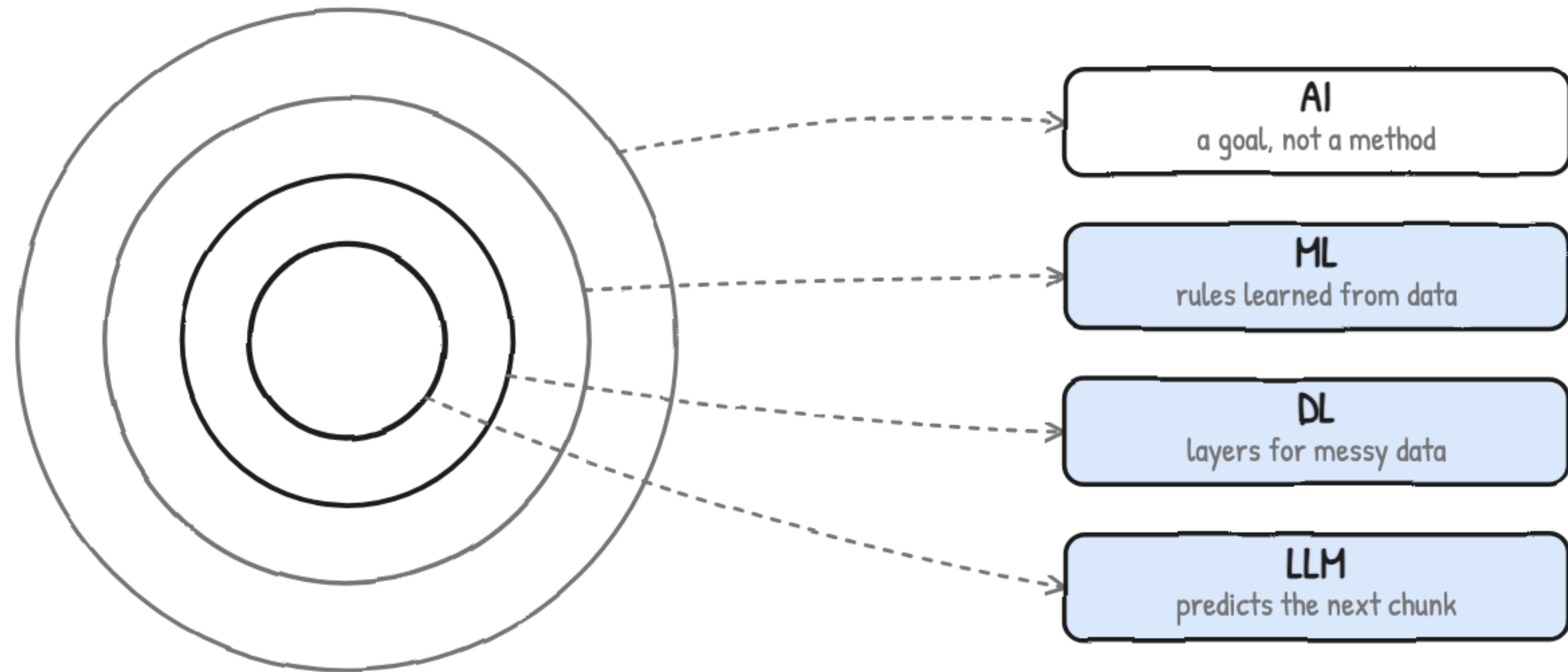
at scale

answers questions
writes code
holds a conversation

trained at internet scale

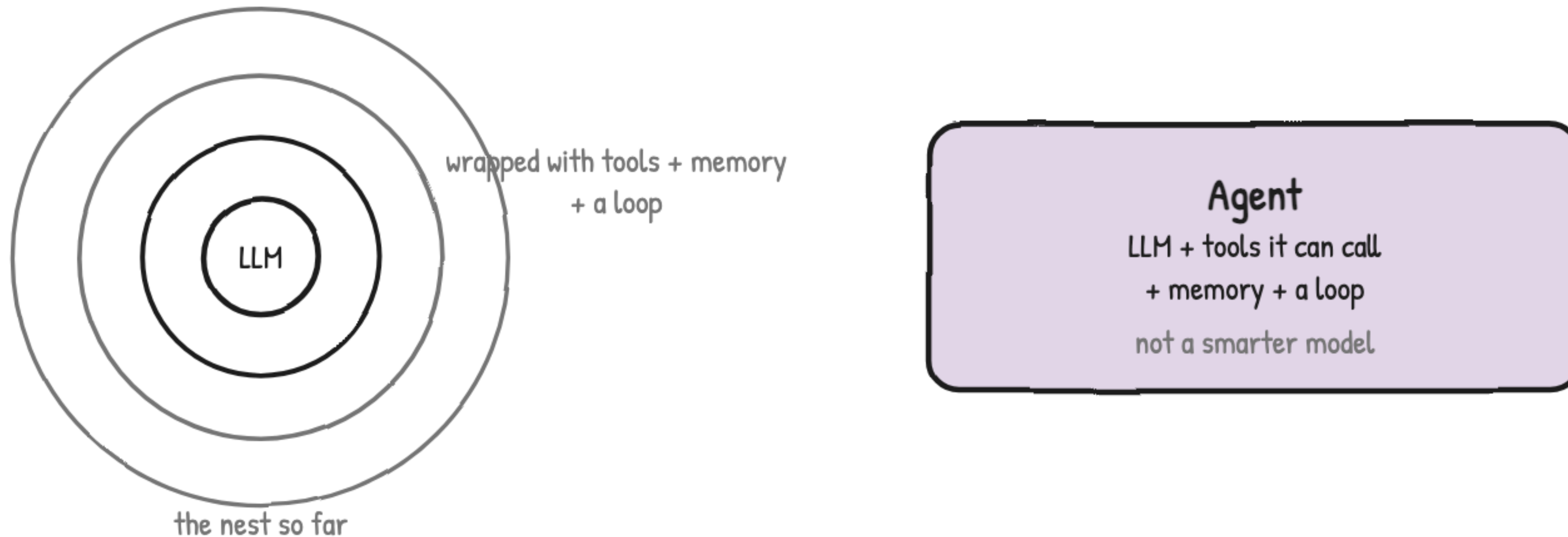
Predict the next chunk of text

Four rings, one nested map



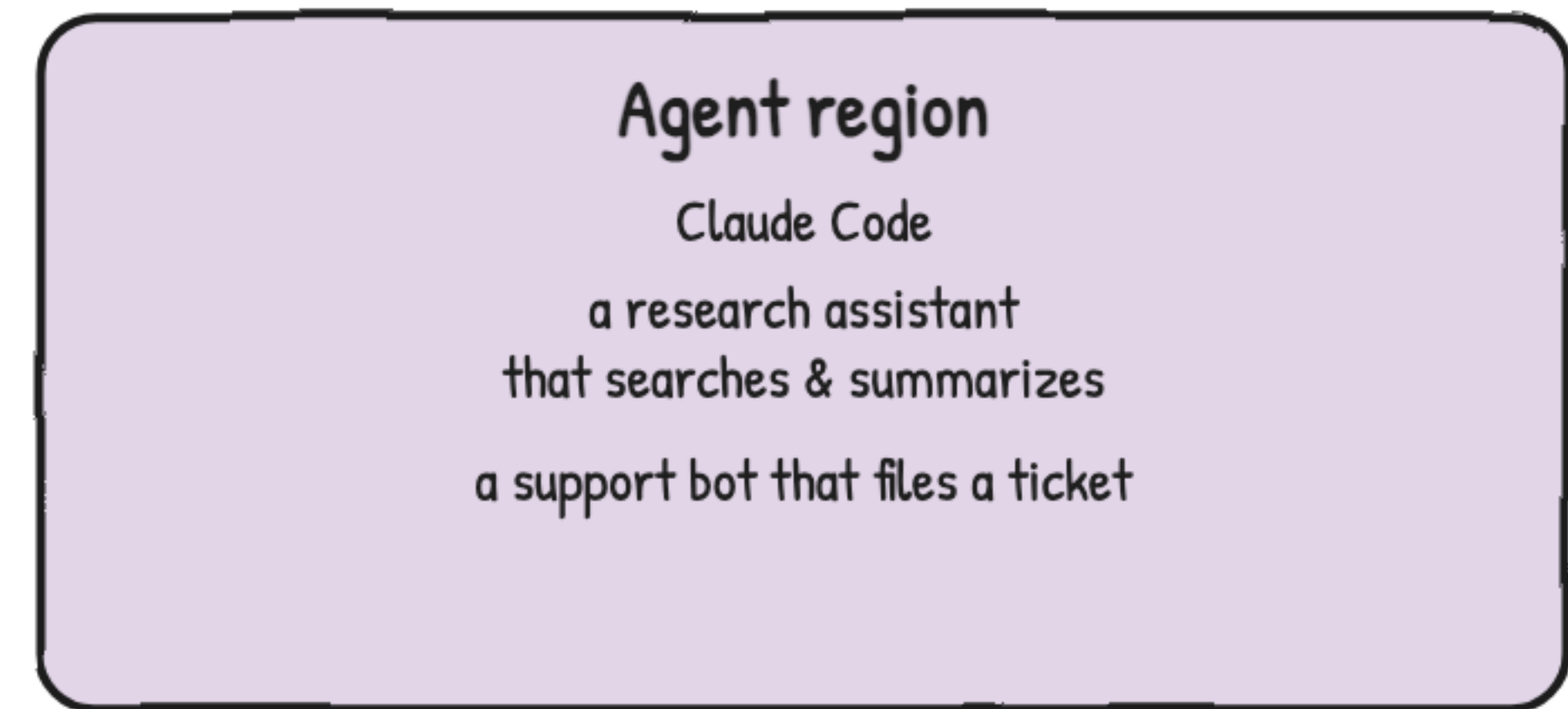
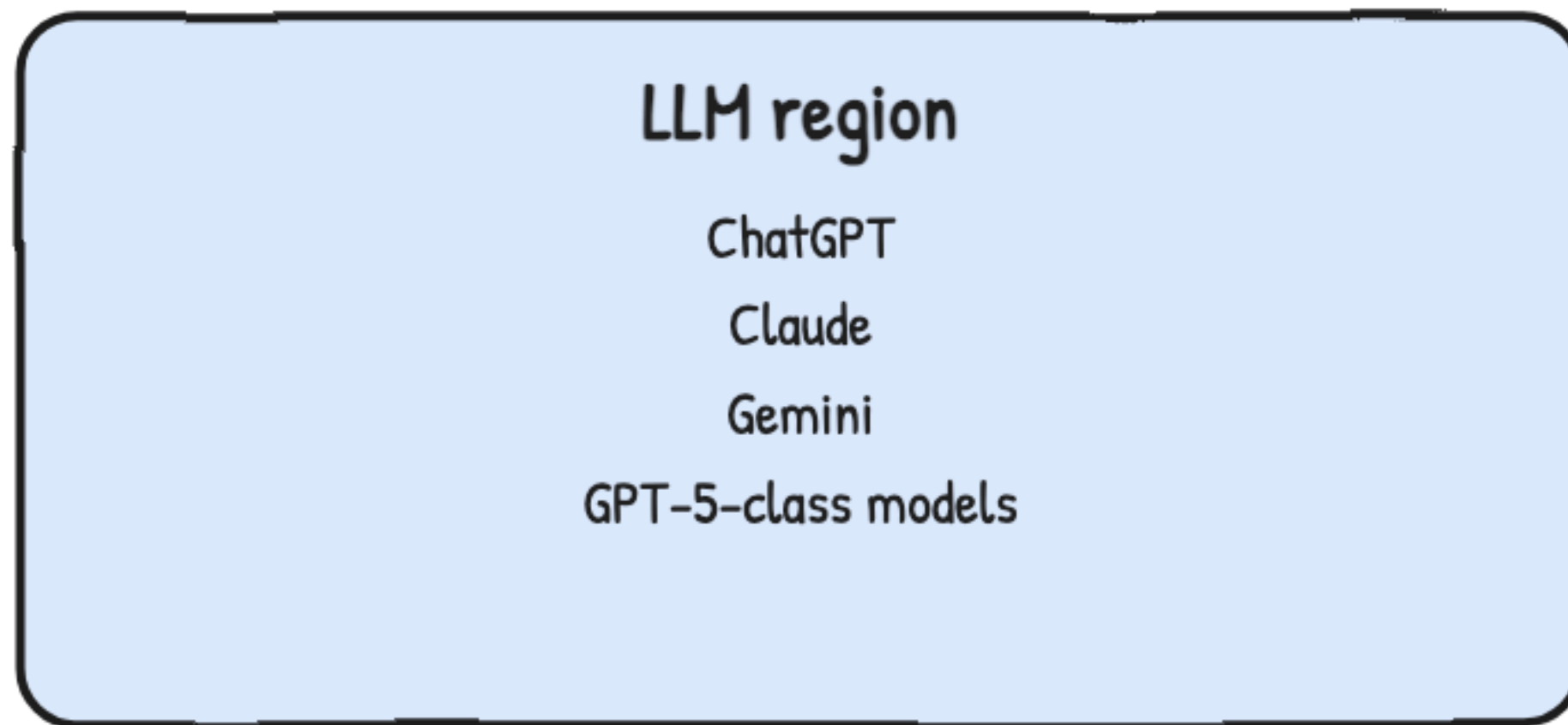
Not four competing technologies

Agents wrap outward, they don't nest inward



Not a smarter model — an architecture

Same model, different architecture, different region



same model underneath, different wrapping

What wraps the model decides the region

The region you're in decides the question you ask

- 1 ML / DL region → "how was it trained, on what data?"
- 2 LLM region → "how many tokens, what's the context window?"
- 3 Agent region → "what tools, what if it calls the wrong one?"

Ask an ML question about an agent, miss the risk



The vendor-call test

Same word "AI," three completely different pitches

ask the same generic question of all three

1

"AI-powered anomaly detection" → ML

ask: training data, retraining cadence, false-positive rate

2

"an LLM copilot" → LLM

ask: context window, hallucination rate

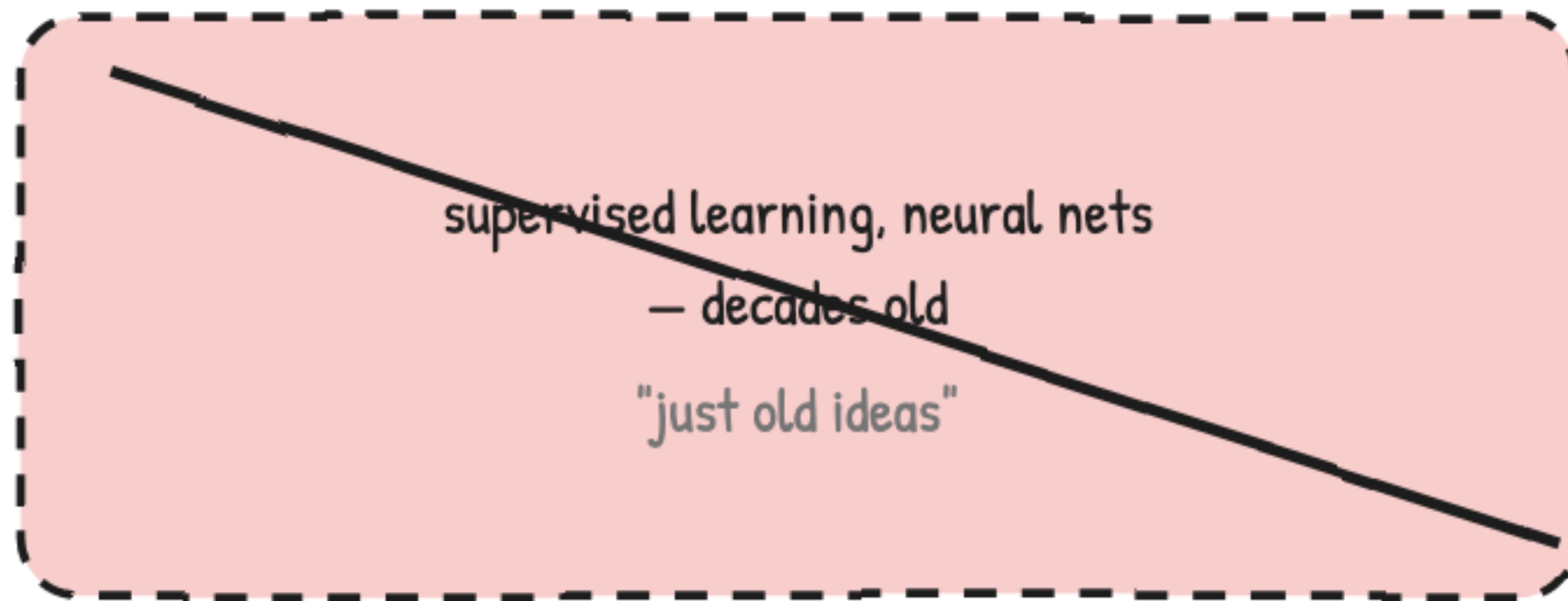
3

"an autonomous agent" → Agent

ask: what tools, who's accountable

Placing the pitch tells you what to ask

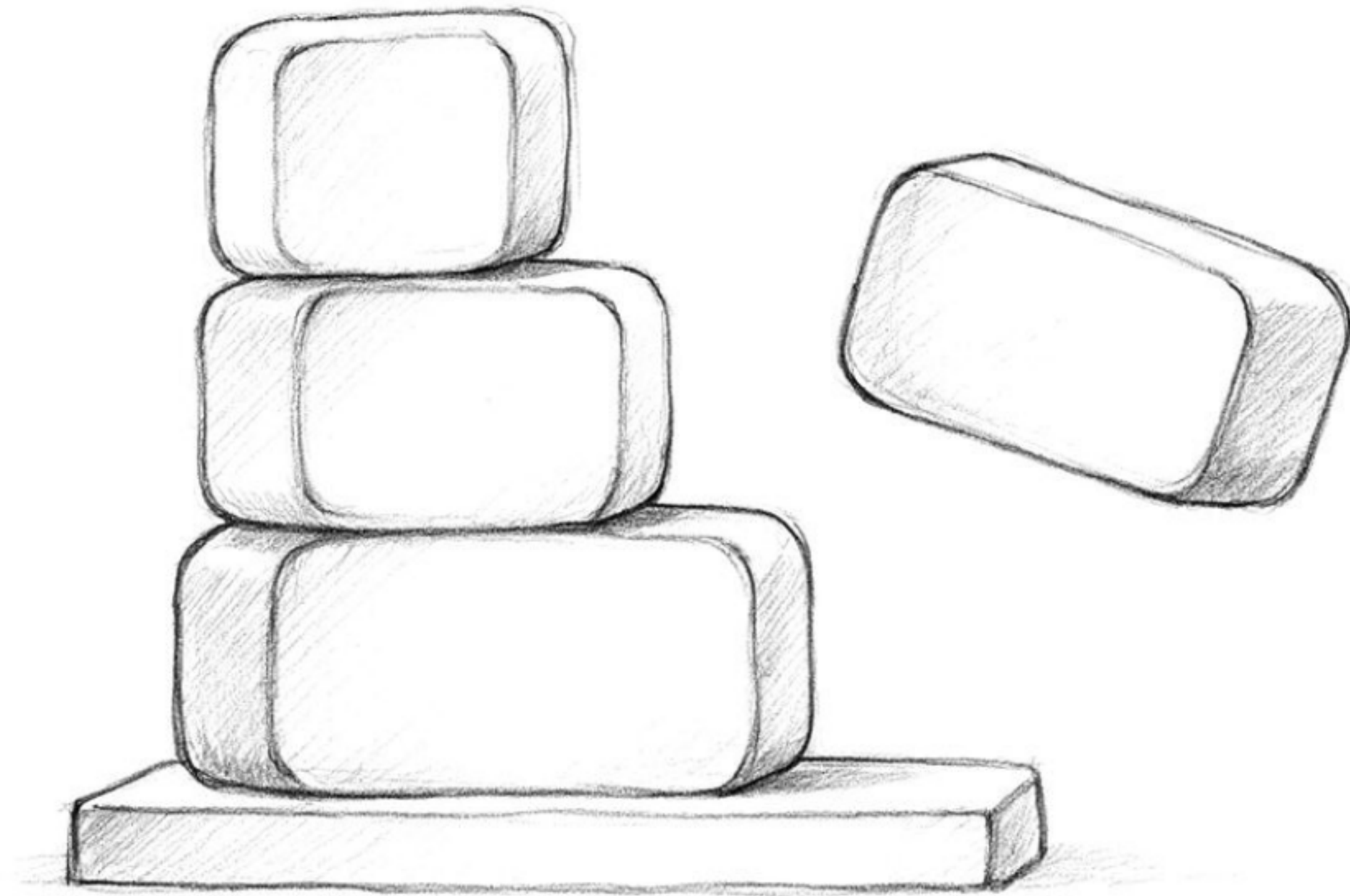
Old ideas, new overlap



but



All four now share one product



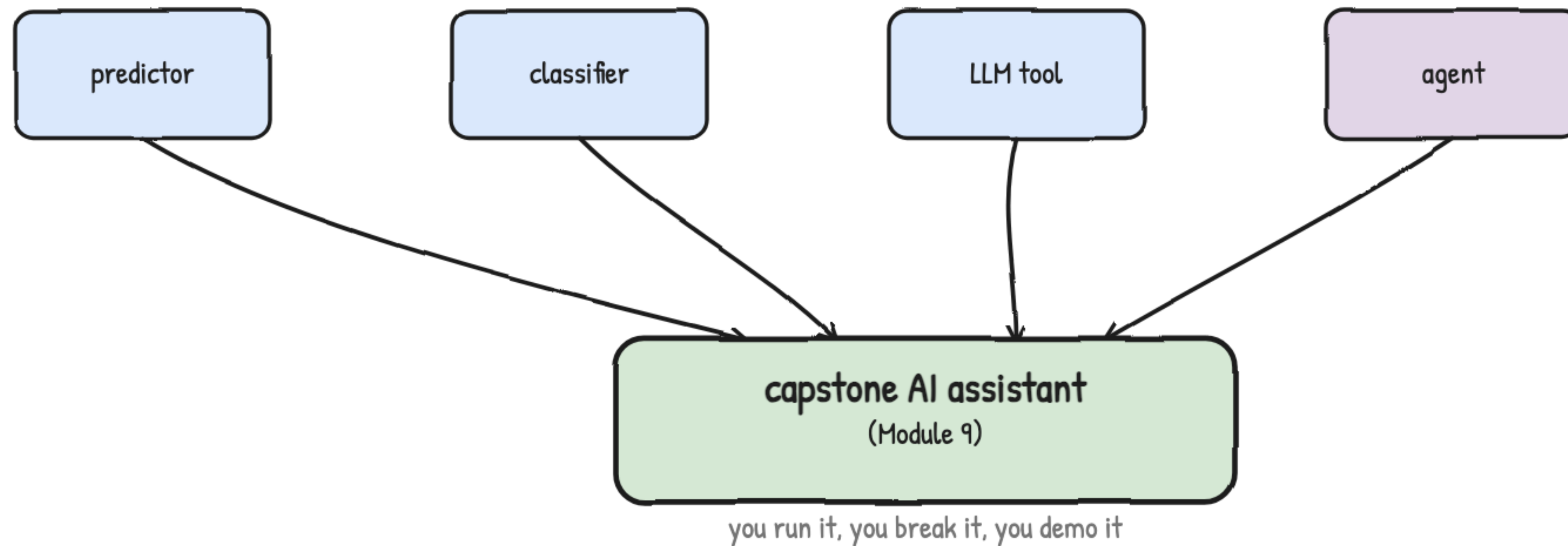
The four projects you'll build

Four projects, four regions, one growing build

- 1 A predictor** (M3)
classical ML — forecasts a number from history
- 2 A digit classifier** (M4)
deep learning — a small neural network
- 3 An LLM-powered tool** (M5)
your first real API call
- 4 An autonomous agent** (M6)
reasons in steps, calls tools

Each project sits in a different region

By the capstone, all four converge into one assistant



You run it, you break it, you demo it

You can place ChatGPT — now make your first call

- 1 AI = a goal, not a method
- 2 ML = rules learned from data
- 3 DL = layered networks for messy data
- 4 LLM = predicts the next chunk, at scale
- 5 Agent = LLM + tools + memory + loop



The map holds; Module 2 gives you the words

Now open **Module 1 · Lab.** · Gourav Shah · School of DevOps & AI